Master 1, Industrial Engineer in Computer Science

Scientific Programming

Will Season 2 of the Netflix Series "One Piece" Be a Success?

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Version of June 1, 2026

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1 Introduction

The objective of this project is to determine whether the upcoming season of the One Piece live action series, produced by Netflix, will be a success, through the production of a time series model.

To achieve this, it will be necessary to define what constitutes a real success, to have a thorough understanding of the dataset used, and then to choose the appropriate tools to create a prediction model.

2 Programming Languages and Modelling of the Prediction Tool

First, the success threshold must be determined, and the dataset must be cleaned and analysed. This involves exploratory data analysis as well as a series of descriptive statistics. This first part will be developed in R. The advantage of this language is that it allows the creation of charts, histograms, and correlograms natively in a single line; the same applies to descriptive statistics.

Once that is done, the bulk of the work consists in modelling a time series to predict the success of the series. This part, which constitutes the majority of the project, will be carried out in Julia. This language has the advantage of being more performant than R; it allows statistical tests to be performed easily via the HypothesisTests library. Complex prediction models can also be easily modelled with it, as will be detailed later. Finally, I feel more confident working on this part in Julia, as it is the language I master best of the two.

3 Definition of Success

The goal of the project is to determine whether the series will be a success; however, the precise condition for success is left to our discretion. I decided to base it on the per-episode rating given by the community on the IMDb website. I defined the threshold as the 3rd quartile of per-episode ratings. If the average rating for the produced season exceeds this threshold, the new season will be considered a success.

After running the R script, the result obtained for the 3rd quartile is **8.1**.

4 Data Preparation

We initially had 8 datasets from the IMDb website provided for model training and final prediction, but these were not up to date and did not contain data on the One Piece live action series. I ultimately used the following dataframes: https://datasets.imdbws.com/. These are in fact the same data, but updated.

The main dataset, *title.ratings.tsv.gz*, was an aggregate of raw data containing per-episode ratings for series, but also mixed overall ratings. I therefore performed a join on the episode ID between this dataset and *title.episode.tsv.gz*, which then allowed me to filter the data and retrieve only the per-episode ratings used for the threshold calculation. Later, I filtered the data to retain only those relating to the One Piece series.

The following table contains the data necessary for the time series prediction:

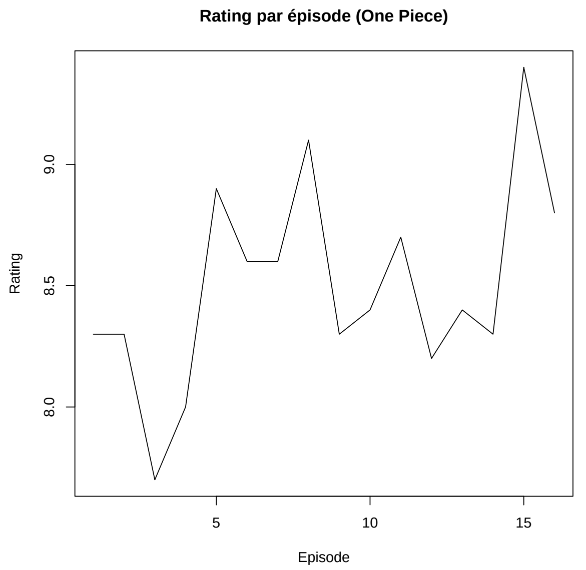
- **tconst**: IMDb episode ID
- **averageRating**: per-episode rating
- **globalEpisode**: global episode number

tconst	averageRating	globalEpisode
tt11748904	8.3	1
tt11758442	8.3	2
tt11758446	7.7	3
tt11758452	8.0	4
tt11758462	8.9	5
tt11758464	8.6	6
tt11758476	8.6	7
tt11758482	9.1	8
tt29137511	8.3	9
tt38079830	8.4	10
tt38079831	8.7	11
tt38079833	8.2	12
tt38079834	8.4	13
tt38079837	8.3	14
tt38079840	9.4	15
tt38752902	8.8	16

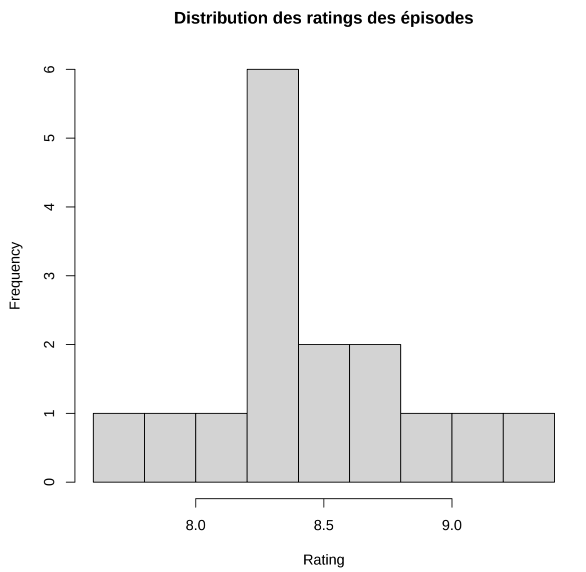
It can be seen that there are only 16 episodes currently, which is a fairly small sample for computing a reliable prediction and training a model. One may therefore question the quality of the final prediction.

5 Exploratory Data Analysis

5.1 Per-Episode Rating Chart

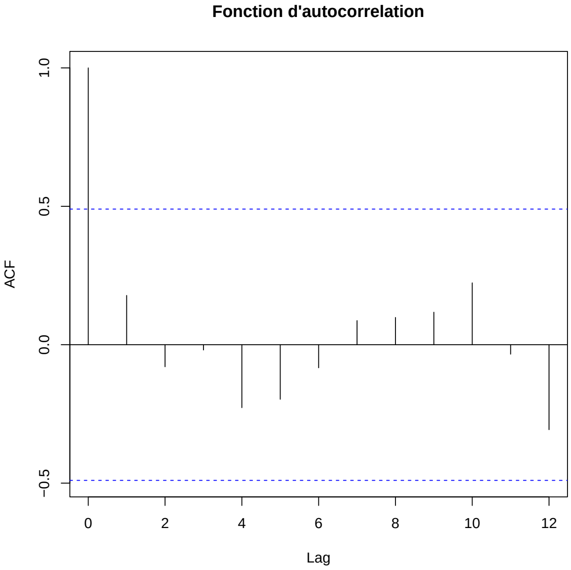


5.2 Histogram of Per-Episode Ratings



Looking at this 9-bar histogram, one can observe that the majority of observations appear to be grouped around the mean, with few data points at the extremes, suggesting a normal distribution. However, this cannot be confirmed with certainty, as the small number of data points does not truly allow the characteristic bell-shaped curve of Gaussian distributions to be observed.

5.3 Correlogram of Ratings



For the autocorrelation function, a large confidence interval of 0.5 is observed. Because of this, no autocorrelation can be declared significantly different from 0 with certainty. This is due to the small observation sample; it can therefore be said that the observed autocorrelation values differ from the true autocorrelations.

5.4 Observations

A clear limitation due to the lack of observational data is apparent, which could have an impact on model training and selection. One possible strategy would be to train a model on a larger set of data in order to obtain more precise observations and make a more informed choice, and only perform the final prediction on the One Piece dataframe. However, this licence occupies a rather unique position, with an already successful manga and animated series, meaning that the way it is rated by fans may differ from traditional series. This is why I nonetheless chose to train a model on the live action data.

6 Descriptive Statistics

Descriptive statistics table for per-episode ratings:

Mean	8.5
Median	8.4
Standard deviation	0.4195235
Maximum	9.4
Minimum	7.7

The ratings are observed to be quite close to one another, with a low standard deviation and therefore low variation across these first two seasons. It can also be seen that the series is already fairly well received by fans, with both the mean and median already exceeding the threshold.

7 Structural Analysis of the Series

Looking at the two preceding points, the series appears to evolve with a seemingly stable trend, around a mean, with relatively low noise. No real seasonality is observed, although this is difficult to assess over only 16 data points. Even though no lags are observed in the correlogram, a lag of 1 or 2 can be inferred given the nature of the observed data (if one episode is well received, the next one likely will be too, and vice versa).

This initial observation might suggest that an ARIMA model would be appropriate for predicting this time series. To verify this, a series of statistical tests will be conducted.

8 Statistical Tests

With the aim of determining the best model, I chose to conduct ADF and Ljung-Box tests. The first is used to determine whether a time series is stable and stationary, while the second tests for autocorrelation. The following results were obtained via the Julia script:

ADF Test

Number of lags	ADF Statistic	p-value	Conclusion
1	0.4324	0.8089	Non-stationary
1	-2.0869	0.2498	Non-stationary
1	-2.5509	0.3030	Non-stationary
2	-1.9087	0.3280	Non-stationary
4	-1.8834	0.3399	Non-stationary

Based on the observed results, all p-values are above 0.05. The results suggest that the series is non-stationary. However, this conclusion must be tempered given the very small number of observations.

Ljung-Box Test

Lag	Q Statistic	p-value	Conclusion
2	0.7387	0.6912	No autocorrelation
4	1.9863	0.7383	No autocorrelation
8	3.7942	0.8752	No autocorrelation

For the Ljung-Box test, the results obtained for lags of 2, 4, and 8 all present a p-value above 0.05. No significant autocorrelation is identified by this test. Once again, this could be due to the low volume of data.

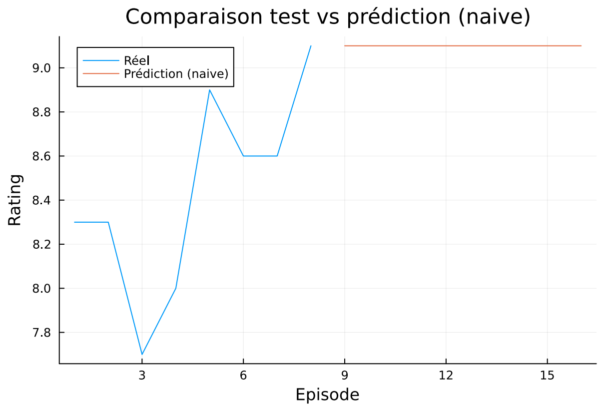
9 Time Series Modelling

For this model training step, I chose to split the dataset into 50% for training and 50% for testing.

First, I modelled the simplest models: naive forecasting, mean forecasting, and drift. Only these three, since the series does not appear to exhibit seasonality. I then modelled ARIMA with different parameters, with the aim of comparing them to determine which is the most performant.

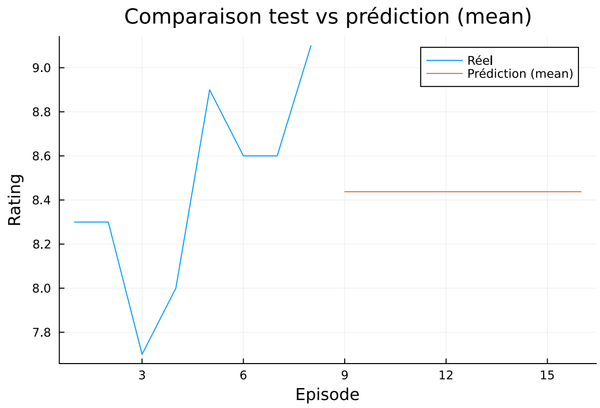
9.1 Simple Predictions

9.1.1 Naive Forecasting



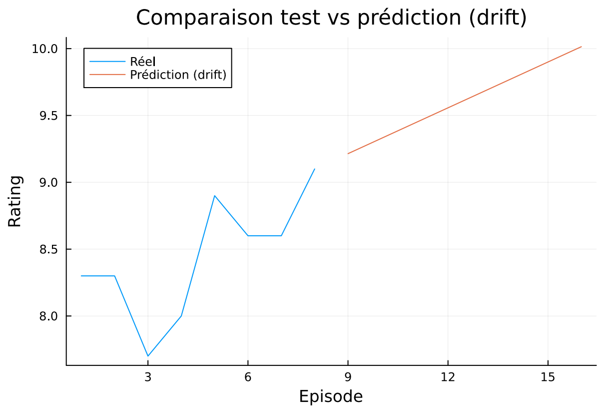
The prediction gives ratings of 9 for the 8 episodes, which is the last value in the training dataset.

9.1.2 Average Forecasting



It can be seen here that the prediction line is close to the mean across the entire dataset. This version therefore appears much better than naive forecasting, which is more suited to cases of an upward or downward trend, which is not the case here.

9.1.3 Drift Forecasting



This model is based on computing each point by adding to the previous point the value of the average slope of the training dataset.

This model is the least suitable of the basic predictions, as it is not stationary at all; the last result even exceeds 10.

9.2 ARIMA

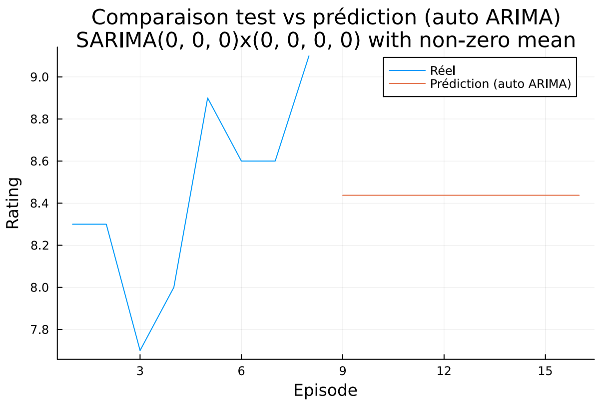
ARIMA is a more advanced prediction tool than the previous models; it is composed of three sub-models:

- **AR**: this is the autoregressive part of the model, accounting for the correlation between a value and the value preceding it.
- **I**: ARIMA is a model suited to predicting stationary time series, which is why the I (integrated) component exists, allowing the trend to be removed in order to make a non-stationary series stationary.
- **MA**: Moving Average. Where AR is based on previous predictions to forecast the series, MA is based on previous errors.

Based on this description and the data already analysed regarding trend, seasonality, and noise, the ARIMA(1,0,0) model appears to be the most appropriate; however, with the aim of obtaining the best-performing model possible, several will be tested.

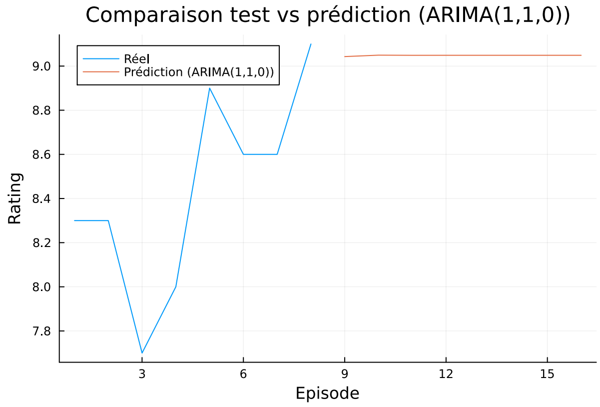
The StateSpaceModels library makes it easy to use the SARIMA model for time series prediction in Julia. This model allows, in addition to the classic ARIMA arguments, the addition of three other factors defining the seasonality of the series. By setting them to 0, the standard ARIMA can be modelled.

9.2.1 Auto ARIMA

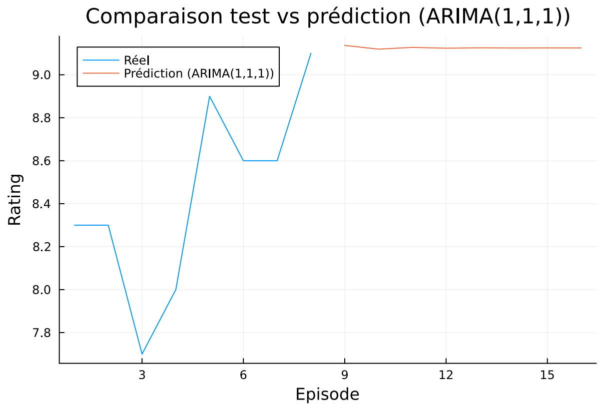


First, I used Auto-ARIMA, an algorithm that calculates the ARIMA parameters it deems most appropriate. The algorithm computed an ARIMA(1,0,0), which confirms my expectation that it would be considered the optimal model. Its predictions are similar, indeed nearly identical, to those obtained by mean prediction.

9.2.2 ARIMA(1,1,0)



9.2.3 ARIMA(1,1,1)



10 Model Evaluation

For this step, I will compare the predicted data against the actual data for each model, using two metrics:

- **MAE**: evaluates the overall average error — i.e., which model makes the fewest errors overall.
- **RMSE**: highlights specifically large individual errors.

The model producing the lowest scores for both metrics will therefore be considered the most performant.

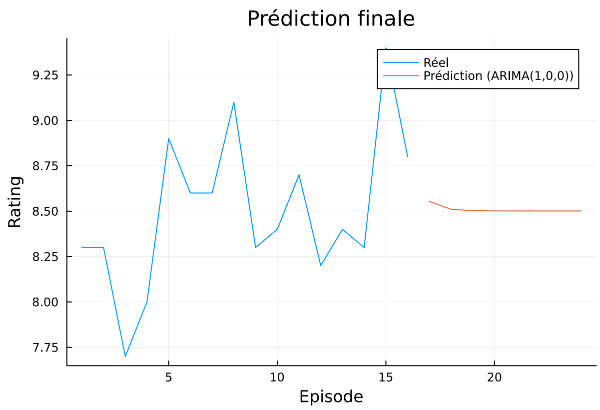
Model	MAE	RMSE
Naive	0.6125	0.6529
Mean	0.2719	0.3911
Drift	1.0518	1.0977
Auto-ARIMA	0.2719	0.3911
ARIMA(1,1,0)	0.3900	0.4355
ARIMA(1,1,1)	0.3900	0.4355

It is clearly apparent that mean prediction and ARIMA(1,0,0) (Auto-ARIMA) have exactly the same results, and are also the best-performing models. Here, I will choose

ARIMA(1,0,0) as the final prediction model, as it offers a more general statistical modelling approach and is potentially more robust on larger series than mean prediction.

11 Predictions

I therefore decided to predict the next 8 episodes, corresponding to season 3 (8 episodes in season 1, 8 episodes in season 2), using the ARIMA(1,0,0) model.



globalEpisode	averageRating
17	8.5538
18	8.5105
19	8.5029
20	8.5016
21	8.5013
22	8.5013
23	8.5013
24	8.5013

This gives us an average per-episode rating of **8.51**, which exceeds the threshold of 8.1 initially set. It can therefore be concluded that, according to the model used, season 3 of the One Piece series will indeed be a success.

12 Conclusion

In conclusion, this document traces the process that allowed the potential success of season 3 of the One Piece live action to be predicted, through the training and testing of various models, ultimately leading to the selection of the ARIMA(1,0,0) model for an optimal final prediction.

This has allowed us to determine that, according to the prediction method used, the next season of the One Piece live action series on Netflix will indeed be a success.

However, the observed results must be nuanced by acknowledging the limitations of the approach. The dataset used as the basis for modelling contained only data from the first seasons, i.e. 16 data points, which inevitably had a significant impact on prediction quality. Furthermore, the prediction process took into account only a single parameter, whereas the popularity of this type of work would appear to be influenced by a whole series of external factors. In addition, the data used — per-episode ratings — were probably not the best possible success indicator (they reflect appreciation, but not the number of people interested). Finally, the threshold estimated for success was defined to give a conclusion to the report, but does not constitute a real success indicator.

Keywords: Time series, Julia, R, ARIMA, Prediction

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